

Second Section.

Theta Functions.

§ 22. The addition theorem.

If u, v are two variables, then the products

$$\vartheta_{g_1, g_2}(u+v)\vartheta_{g'_1, g'_2}(u-v)$$

belong to the Θ -functions of the second order, with characteristic

$$(g_1 + g'_1, g_2 + g'_2),$$

and indeed for each of the two variables u, v ; regarded as functions of v they are therefore linearly representable by the functions (12) and (15) of § 21, so that the variable u occurs in the coefficients. When the characteristic is $(0, 0)$ this representation is possible in several ways, since one can choose two arbitrary functions from (12); in the other cases it is possible in only one way. One obtains in this manner sixteen formulae, of which, however, six can be derived from the others by mere interchange of v with $-v$, so that only ten essentially different formulae remain. These formulae are very easily derived by first writing them with undetermined coefficients and then determining these coefficients by putting for v such special values for which one of the functions $\vartheta(v)$ vanishes. Thus, for example,

$$\vartheta_{00}(u+v)\vartheta_{00}(u-v) = A\vartheta_{00}^2(v) + B\vartheta_{11}^2(v),$$

and if one puts $v = 0$ and $v = (1 + \omega)/2$, one obtains, using the formulae of § 21,

$$A\vartheta_{00}^2 = \vartheta_{00}^2(u), \quad B\vartheta_{00}^2 = \vartheta_{11}^2(u),$$

and hence

$$\vartheta_{00}^2\vartheta_{00}(u+v)\vartheta_{00}(u-v) = \vartheta_{00}^2(u)\vartheta_{00}^2(v) + \vartheta_{11}^2(u)\vartheta_{11}^2(v). \quad (1)$$

If one replaces u in this by

$$u + \frac{1}{2}, \quad u + \frac{\omega}{2}, \quad u + \frac{1+\omega}{2},$$

one obtains three further relations, which could also have been derived directly in the same way as (1):

$$\vartheta_{11}^2\vartheta_{01}(u+v)\vartheta_{01}(u-v) = \vartheta_{01}^2(u)\vartheta_{01}^2(v) - \vartheta_{11}^2(u)\vartheta_{11}^2(v), \quad (2)$$

$$\vartheta_{11}^2\vartheta_{10}(u+v)\vartheta_{10}(u-v) = \vartheta_{10}^2(u)\vartheta_{10}^2(v) - \vartheta_{11}^2(u)\vartheta_{11}^2(v), \quad (3)$$

$$\vartheta_{01}^2\vartheta_{11}(u+v)\vartheta_{11}(u-v) = \vartheta_{11}^2(u)\vartheta_{01}^2(v) - \vartheta_{01}^2(u)\vartheta_{11}^2(v). \quad (4)$$

These four equations can, as already mentioned, be transformed in many ways by applying the formulae (13) of the preceding paragraph. The following six formulae have only one form. It is, to begin again with an arbitrary example,

$$\vartheta_{00}(u+v)\vartheta_{11}(u-v) = A\vartheta_{01}(v)\vartheta_{10}(v) + B\vartheta_{00}(v)\vartheta_{11}(v),$$

and if one puts $v = 0$,

$$A\vartheta_{01}\vartheta_{10} = \vartheta_{00}(u)\vartheta_{11}(u);$$

from this one obtains B by interchanging u with v . Thus the three following formulae are derived:

$$\vartheta_{01}\vartheta_{10}\vartheta_{00}(u+v)\vartheta_{11}(u-v) = \vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{10}(v) - \vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{11}(v), \quad (5)$$

$$\vartheta_{00}\vartheta_{01}\vartheta_{10}(u+v)\vartheta_{11}(u-v) = \vartheta_{10}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{01}(v) - \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{10}(v)\vartheta_{11}(v), \quad (6)$$

$$\vartheta_{00}\vartheta_{10}\vartheta_{01}(u+v)\vartheta_{11}(u-v) = \vartheta_{01}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{10}(v) - \vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{01}(v)\vartheta_{11}(v), \quad (7)$$

which are verified immediately by putting first u , then v , equal to zero. From these the three others follow by increasing u by

$$\frac{1}{2}, \quad \frac{\omega}{2}, \quad \frac{1}{2} :$$

$$\vartheta_{01}\vartheta_{10}\vartheta_{01}(u+v)\vartheta_{10}(u-v) = \vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{01}(v)\vartheta_{10}(v) + \vartheta_{00}(u)\vartheta_{11}(u)\vartheta_{00}(v)\vartheta_{11}(v), \quad (8)$$

$$\vartheta_{00}\vartheta_{01}\vartheta_{00}(u+v)\vartheta_{01}(u-v) = \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{00}(v)\vartheta_{01}(v) - \vartheta_{10}(u)\vartheta_{11}(u)\vartheta_{10}(v)\vartheta_{11}(v), \quad (9)$$

$$\vartheta_{00}\vartheta_{10}\vartheta_{00}(u+v)\vartheta_{10}(u-v) = \vartheta_{00}(u)\vartheta_{10}(u)\vartheta_{00}(v)\vartheta_{10}(v) + \vartheta_{01}(u)\vartheta_{11}(u)\vartheta_{01}(v)\vartheta_{11}(v). \quad (10)$$

The formulae whose totality is designated by the name of the addition theorem can be generalized in many ways; the following may serve as an example.

If u, v are two variables, then the four products

$$\vartheta_{00}(u)\vartheta_{00}(u+v), \quad \vartheta_{01}(u)\vartheta_{01}(u+v), \quad \vartheta_{10}(u)\vartheta_{10}(u+v), \quad \vartheta_{11}(u)\vartheta_{11}(u+v),$$

regarded as functions of u , are related T -functions of the second order with the same periodicity factors, and hence any three of them are linearly dependent. Thus

$$A\vartheta_{00}(u)\vartheta_{00}(u+v) + B\vartheta_{10}(u)\vartheta_{10}(u+v) + C\vartheta_{11}(u)\vartheta_{11}(u+v) = 0,$$

from which, for $u = 0, u = \frac{1}{2}$,

$$A\vartheta_{00}\vartheta_{00}(v) + B\vartheta_{10}\vartheta_{10}(v) = 0, \quad A\vartheta_{01}\vartheta_{01}(v) + C\vartheta_{10}\vartheta_{10}(v) = 0,$$

and hence

$$\vartheta_{10}\vartheta_{10}(v)\vartheta_{00}(u)\vartheta_{00}(u+v) - \vartheta_{00}\vartheta_{00}(v)\vartheta_{10}(u)\vartheta_{10}(u+v) - \vartheta_{01}\vartheta_{01}(v)\vartheta_{11}(u)\vartheta_{11}(u+v) = 0. \quad (11)$$

Likewise, if w is a third variable, the products contained in the form

$$\vartheta_{g_1, g_2}(u+v)\vartheta_{g_1, g_2}(u+w), \quad \vartheta_{g_1, g_2}(u)\vartheta_{g_1, g_2}(u+v+w),$$

regarded as functions of u , are related T -functions of the second order with the same periodicity factors, so that a linear dependence exists between any three among them. The coefficients are determined, as above, by special values of u . Thus follows the system of formulae:

$$\begin{aligned} \vartheta_{00}\vartheta_{00}(v+w)\vartheta_{00}(u+v)\vartheta_{00}(u+w) &= \vartheta_{00}(u)\vartheta_{00}(v)\vartheta_{00}(w)\vartheta_{00}(u+v+w) \\ &\quad - \vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w), \end{aligned} \quad (12)$$

$$\begin{aligned} \vartheta_{01}\vartheta_{01}(v+w)\vartheta_{01}(u+v)\vartheta_{01}(u+w) &= \vartheta_{01}(u)\vartheta_{01}(v)\vartheta_{01}(w)\vartheta_{01}(u+v+w) \\ &\quad + \vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w), \end{aligned} \quad (13)$$

$$\begin{aligned} \vartheta_{10}\vartheta_{10}(v+w)\vartheta_{10}(u+v)\vartheta_{10}(u+w) &= \vartheta_{10}(u)\vartheta_{10}(v)\vartheta_{10}(w)\vartheta_{10}(u+v+w) \\ &\quad + \vartheta_{11}(u)\vartheta_{11}(v)\vartheta_{11}(w)\vartheta_{11}(u+v+w). \end{aligned} \quad (14)$$

All these formulae can be regarded as special cases of a general formula which Jacobi derived from the series expansions and used for the foundation of the theory of elliptic functions.¹ This formula can also be obtained from the concept of theta functions as follows.

The four functions

$$\vartheta_{00}(2v), \quad \vartheta_{01}(2v), \quad \vartheta_{10}(2v), \quad \vartheta_{11}(2v) \quad (15)$$

are Θ_{00} -functions of the fourth order in v , and since they are linearly independent, all Θ_{00} -functions of the fourth order can be expressed linearly by them. But the product

$$\vartheta(v+a_1)\vartheta(v+a_2)\vartheta(v+a_3)\vartheta(v+a_4) \quad (16)$$

¹Theory of elliptic functions, derived from the properties of theta series. Jacobi's collected works, vol. I, p. 497.

is also such a function, provided that the quantities a satisfy the condition

$$a_1 + a_2 + a_3 + a_4 = 0. \quad (17)$$

Thus, denoting by A_1, A_2, A_3, A_4 constants with respect to v , we obtain

$$\vartheta(v + a_1)\vartheta(v + a_2)\vartheta(v + a_3)\vartheta(v + a_4) = A_1\vartheta_{00}(2v) + A_2\vartheta_{01}(2v) + A_3\vartheta_{10}(2v) + A_4\vartheta_{11}(2v).$$

From this, by increasing v by $\frac{1}{2}(g_1\omega + g_2)$, one obtains four formulae:

$$\begin{aligned} & \vartheta_{g_1, g_2}(v + a_1)\vartheta_{g_1, g_2}(v + a_2)\vartheta_{g_1, g_2}(v + a_3)\vartheta_{g_1, g_2}(v + a_4) \\ &= A_1\vartheta_{00}(2v) + (-1)^{g_2}A_2\vartheta_{01}(2v) + (-1)^{g_1}A_3\vartheta_{10}(2v) + (-1)^{g_1+g_2}A_4\vartheta_{11}(2v). \end{aligned}$$

If one adds these four formulae, then

$$4A_1\vartheta_{00}(2v) = \sum_{g_1, g_2} \vartheta_{g_1, g_2}(v + a_1)\vartheta_{g_1, g_2}(v + a_2)\vartheta_{g_1, g_2}(v + a_3)\vartheta_{g_1, g_2}(v + a_4), \quad (18)$$

where in the sum all combinations of 0 and 1 are to be put for g_1 and g_2 .

We now introduce a slightly different notation by putting

$$v + a_1 = v'_1, \quad v + a_2 = v'_2, \quad v + a_3 = v'_3, \quad v + a_4 = v'_4,$$

so that, because of (17),

$$4v = v'_1 + v'_2 + v'_3 + v'_4,$$

and now define the four quantities v_1, v_2, v_3, v_4 by the equations

$$\begin{aligned} v_1 &= \frac{1}{2}(v'_1 + v'_2 + v'_3 + v'_4), & v_2 &= \frac{1}{2}(v'_1 + v'_2 - v'_3 - v'_4), \\ v_3 &= \frac{1}{2}(v'_1 - v'_2 + v'_3 - v'_4), & v_4 &= \frac{1}{2}(v'_1 - v'_2 - v'_3 + v'_4). \end{aligned} \quad (19)$$

From this one obtains conversely

$$\begin{aligned} v'_1 &= \frac{1}{2}(v_1 + v_2 + v_3 + v_4), & a_1 &= \frac{1}{2}(v_2 + v_3 + v_4), \\ v'_2 &= \frac{1}{2}(v_1 + v_2 - v_3 - v_4), & a_2 &= \frac{1}{2}(v_2 - v_3 - v_4), \\ v'_3 &= \frac{1}{2}(v_1 - v_2 + v_3 - v_4), & a_3 &= \frac{1}{2}(-v_2 + v_3 - v_4), \\ v'_4 &= \frac{1}{2}(v_1 - v_2 - v_3 + v_4), & a_4 &= \frac{1}{2}(-v_2 - v_3 + v_4), \end{aligned} \quad (20)$$

and thereafter formula (18) gives

$$4A_1\vartheta_{00}(v_1) = \sum_{g_1, g_2} \vartheta_{g_1, g_2}(v'_1)\vartheta_{g_1, g_2}(v'_2)\vartheta_{g_1, g_2}(v'_3)\vartheta_{g_1, g_2}(v'_4),$$

where now either the v_i or the v'_i can be regarded as independent variables.

If the v_i are regarded as independent variables, then in this formula A_1 is independent of v_1 , but still depends on v_2, v_3, v_4 . Thus put

$$4A_1 = c\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4).$$

Then

$$c\vartheta_{00}(v_1)\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4) = \sum \vartheta_{g_1, g_2}(v'_1)\vartheta_{g_1, g_2}(v'_2)\vartheta_{g_1, g_2}(v'_3)\vartheta_{g_1, g_2}(v'_4),$$

and here c is in any case independent of v_1 . But since the right hand side of this formula remains unchanged under arbitrary interchanges of v_1, v_2, v_3, v_4 , c is independent of all v_i ; and its value is obtained by putting all $v_i = 0$:

$$c\vartheta_{00}^4 = \vartheta_{00}^4 + \vartheta_{01}^4 + \vartheta_{10}^4,$$

whence by § 21, (14), $c = 2$ follows. Thus we have the formula

$$2\vartheta_{00}(v_1)\vartheta_{00}(v_2)\vartheta_{00}(v_3)\vartheta_{00}(v_4) = \sum_{g_1, g_2} \vartheta_{g_1, g_2}(v'_1)\vartheta_{g_1, g_2}(v'_2)\vartheta_{g_1, g_2}(v'_3)\vartheta_{g_1, g_2}(v'_4).$$

If in it each of the variables v_1, v_2, v_3, v_4 is increased by the half-period $-\frac{1}{2}(g'_1\omega - g'_2)$, then v'_1 is increased by a whole period $-(g'_1\omega - g'_2)$, while v'_2, v'_3, v'_4 remain unchanged. Accordingly, with the aid of § 21, (2) and (3), one obtains a system of four formulae:

$$2\vartheta_{g'_1, g'_2}(v_1)\vartheta_{g'_1, g'_2}(v_2)\vartheta_{g'_1, g'_2}(v_3)\vartheta_{g'_1, g'_2}(v_4) = \sum_{g_1, g_2} (-1)^{g_1 g'_2 + g_2 g'_1 + g'_1 g'_2} \vartheta_{g_1, g_2}(v'_1)\vartheta_{g_1, g_2}(v'_2)\vartheta_{g_1, g_2}(v'_3)\vartheta_{g_1, g_2}(v'_4). \quad (21)$$

These are Jacobi's formulae.

If, for example, one puts

$$v_1 = 0, \text{quad} v_2 = v + w, \quad v_3 = u + v, \quad v_4 = u + w,$$

$$v'_1 = u + v + w, \quad v'_2 = -u, \quad v'_3 = -w, \quad v'_4 = -v,$$

then from (21), for $g'_1, g'_2 = 1, 1$,

$$0 = \sum_{g_1, g_2} (-1)^{g_1 + g_2} \vartheta_{g_1, g_2}(u)\vartheta_{g_1, g_2}(v)\vartheta_{g_1, g_2}(w)\vartheta_{g_1, g_2}(u + v + w),$$

and from this one obtains, by putting in (21) the three other characteristics for g'_1, g'_2 , the formulae (12), (13), (14).

§ 23. The derivatives of the ϑ -functions.

In what follows we denote by $\vartheta'_{g_1, g_2}(v)$, $\vartheta''_{g_1, g_2}(v)$ the derivatives, taken with respect to v , of the function $\vartheta_{g_1, g_2}(v)$, and by ϑ'_{g_1, g_2} , ϑ''_{g_1, g_2} the values of this function for $v = 0$. Then $\vartheta'_{01}, \vartheta'_{10}, \vartheta'_{00}$ vanish, because $\vartheta_{01}(v), \vartheta_{10}(v), \vartheta_{00}(v)$ are even functions of v , and likewise ϑ_{11} vanishes.

The six functions

$$\vartheta_{g_1, g_2}(v)\vartheta'_{g'_1, g'_2}(v) - \vartheta_{g'_1, g'_2}(v)\vartheta'_{g_1, g_2}(v)$$

are, as follows from the fundamental equations § 20, (1), ϑ -functions of the second order with characteristic

$$(g_1 + g'_1, g_2 + g'_2),$$

and at the same time either even or odd functions; accordingly they can be represented by ϑ -functions according to § 21. A constant factor is determined by a special value of v , namely $v = 0$. Thus

$$\vartheta'_{11}(v)\vartheta_{01}(v) - \vartheta'_{01}(v)\vartheta_{11}(v) = A\vartheta_{10}(v)\vartheta_{00}(v), \quad (1)$$

$$A = \frac{\vartheta'_{11}\vartheta_{01}}{\vartheta_{10}\vartheta_{00}}. \quad (2)$$

But this expression for A can still be simplified. If we differentiate (1) twice with respect to v and then put $v = 0$, we get

$$\vartheta'''_{11}\vartheta_{01} - \vartheta''_{01}\vartheta'_{11} = A(\vartheta''_{10}\vartheta_{00} + \vartheta_{10}\vartheta''_{00}),$$

and if for A we insert the value (2),

$$\frac{\vartheta'''_{11}}{\vartheta'_{11}} = \frac{\vartheta''_{01}}{\vartheta_{01}} + \frac{\vartheta''_{10}}{\vartheta_{10}} + \frac{\vartheta''_{00}}{\vartheta_{00}}. \quad (3)$$

According to § 20, (4), however, the four functions $\vartheta_{11}, \vartheta_{01}, \vartheta_{10}, \vartheta_{00}$ satisfy the differential equation

$$\vartheta'' = 4\pi i \frac{\partial \vartheta}{\partial \omega}, \quad (4)$$

and accordingly (3) can be written

$$\frac{d \log \vartheta'_{11}}{d\omega} = \frac{d \log \vartheta_{00} \vartheta_{10} \vartheta_{01}}{d\omega},$$

or, by integration,

$$\vartheta'_{11} = c \vartheta_{00} \vartheta_{10} \vartheta_{01},$$

where c is independent of ω . By § 20, 21 the ϑ -functions were determined up to a factor independent of v, ω and common to all of them. This factor is now to be chosen so that the constant c obtains the value π ; thus

$$\vartheta'_{11} = \pi \vartheta_{00} \vartheta_{10} \vartheta_{01}, \quad (5)$$

and thereby the ϑ -functions are now completely defined, up to the common sign $\pm$. By application of (5), formula (1) assumes the form

$$\vartheta'_{11}(v) \vartheta_{01}(v) - \vartheta'_{01}(v) \vartheta_{11}(v) = \pi \vartheta_{01}^2 \vartheta_{10}(v) \vartheta_{00}(v), \quad (6)$$

and similarly, using § 21, (8), one obtains

$$\vartheta'_{11}(v) \vartheta_{00}(v) - \vartheta'_{00}(v) \vartheta_{11}(v) = \pi \vartheta_{00}^2 \vartheta_{10}(v) \vartheta_{01}(v), \quad (7)$$

$$\vartheta'_{11}(v) \vartheta_{10}(v) - \vartheta'_{10}(v) \vartheta_{11}(v) = \pi \vartheta_{10}^2 \vartheta_{01}(v) \vartheta_{00}(v), \quad (8)$$

$$\vartheta'_{10}(v) \vartheta_{01}(v) - \vartheta'_{01}(v) \vartheta_{10}(v) = -\pi \vartheta_{00}^2 \vartheta_{00}(v) \vartheta_{11}(v), \quad (9)$$

$$\vartheta'_{00}(v) \vartheta_{01}(v) - \vartheta'_{01}(v) \vartheta_{00}(v) = -\pi \vartheta_{10}^2 \vartheta_{10}(v) \vartheta_{11}(v), \quad (10)$$

$$\vartheta'_{00}(v) \vartheta_{10}(v) - \vartheta'_{10}(v) \vartheta_{00}(v) = \pi \vartheta_{01}^2 \vartheta_{01}(v) \vartheta_{11}(v). \quad (11)$$

If the equations (9), (10), (11) are differentiated with respect to v and then $v = 0$ is put, we obtain by means of (4) and (5) the relations

$$4 \frac{d}{d\omega} \log \frac{\vartheta_{10}}{\vartheta_{01}} = i\pi \vartheta_{00}^4, \quad (12)$$

$$4 \frac{d}{d\omega} \log \frac{\vartheta_{00}}{\vartheta_{01}} = i\pi \vartheta_{10}^4, \quad (13)$$

$$4 \frac{d}{d\omega} \log \frac{\vartheta_{10}}{\vartheta_{00}} = i\pi \vartheta_{01}^4. \quad (14)$$

If the fundamental formulae for the ϑ -functions are differentiated logarithmically twice, one sees that the functions

$$\vartheta^2(v) \frac{d^2 \log \vartheta(v)}{dv^2} = \vartheta''(v) \vartheta(v) - \vartheta'(v) \vartheta'(v)$$

are Θ -functions of the second order with characteristic $(0, 0)$, and can therefore be expressed linearly by the functions $\vartheta^2(v)$. In this way one obtains

$$\vartheta_{00}^2 \vartheta_{00}^2(v) \frac{d^2 \log \vartheta_{00}(v)}{dv^2} = \vartheta_{00} \vartheta_{00}'' \vartheta_{00}^2(v) + \vartheta_{11}^2 \vartheta_{11}^2(v), \quad (15)$$

$$\vartheta_{01}^2 \vartheta_{01}^2(v) \frac{d^2 \log \vartheta_{01}(v)}{dv^2} = \vartheta_{01} \vartheta_{01}'' \vartheta_{01}^2(v) - \vartheta_{11}^2 \vartheta_{11}^2(v), \quad (16)$$

$$\vartheta_{10}^2 \vartheta_{10}^2(v) \frac{d^2 \log \vartheta_{10}(v)}{dv^2} = \vartheta_{10} \vartheta_{10}'' \vartheta_{10}^2(v) - \vartheta_{11}^2 \vartheta_{11}^2(v), \quad (17)$$

$$\vartheta_{00}^2 \vartheta_{11}^2(v) \frac{d^2 \log \vartheta_{11}(v)}{dv^2} = \vartheta_{00} \vartheta_{00}'' \vartheta_{11}^2(v) - \vartheta_{11}^2 \vartheta_{00}^2(v). \quad (18)$$

From this further relations can be derived by continued differentiation. We mention one more of these formulae, which results when one differentiates (15) twice more with respect to v and then puts $v = 0$. Expressing the differentiations with respect to v by such with respect to ω , by means of the partial differential equation (4), one obtains

$$\frac{d^2 \log \vartheta_{00}}{d\omega^2} - 2 \left(\frac{d \log \vartheta_{00}}{d\omega} \right)^2 = -\frac{\pi^2}{8} \vartheta_{10}^4 \vartheta_{01}^4,$$

or

$$\frac{d}{d\omega} \left(\frac{1}{\vartheta_{00}^4} \frac{d\vartheta_{00}^2}{d\omega} \right) = -\frac{\pi^2}{4} \frac{\vartheta_{10}^4 \vartheta_{01}^4}{\vartheta_{00}^2}. \quad (19)$$

§ 24. Representation of the ϑ -functions by infinite products.

The theory of the ϑ -functions has now advanced far enough that their representation is obtained very easily. Thereby the existence of these functions is proved, and the previous considerations obtain their secure ground. Two paths to this goal stand open to us. The first starts from the zeros of the ϑ -functions already known to us and composes infinite products from them.

By § 21, the function $\vartheta_{00}(v)$ vanishes when

$$2v = (2\nu - 1)\omega + (2\mu - 1)$$

where ν, μ are integers, or when

$$e^{\pm 2\pi i v} = -e^{\pi i \omega (2\nu - 1)},$$

where we may now restrict ν to positive numbers. Thus, for abbreviation, put

$$e^{\pi i \omega} = q;$$

then q is a quantity whose absolute value is a proper fraction. The convergent infinite product

$$P(v) = \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1} e^{2\pi i v})(1 + q^{2\nu-1} e^{-2\pi i v})$$

therefore vanishes in all, and only in, those points in which $\vartheta_{00}(v)$ vanishes. Moreover,

$$P(v + 1) = P(v),$$

and

$$P(v + \omega) = \frac{1 + q^{-1} e^{-2\pi i v}}{1 + q e^{2\pi i v}} P(v) = q^{-1} e^{-2\pi i v} P(v).$$

But this is, by § 20, (1), the periodicity property of the function $\vartheta_{00}(v)$, and therefore

$$\vartheta_{00}(v) = Q \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1} e^{2\pi i v})(1 + q^{2\nu-1} e^{-2\pi i v}), \quad (1)$$

where Q is a factor independent of v . From this, by the formulae (8) of § 21, replacing v by

$$v + \frac{1}{2}, \quad v + \frac{\omega}{2}, \quad v + \frac{1 + \omega}{2},$$

one obtains

$$\vartheta_{01}(v) = Q \prod_{\nu=1}^{\infty} (1 - q^{2\nu-1} e^{2\pi i v})(1 - q^{2\nu-1} e^{-2\pi i v}), \quad (2)$$

$$\vartheta_{10}(v) = Q q^{1/4} (e^{\pi i v} + e^{-\pi i v}) \prod_{\nu=1}^{\infty} (1 + q^{2\nu} e^{2\pi i v})(1 + q^{2\nu} e^{-2\pi i v}), \quad (3)$$

$$\vartheta_{11}(v) = -i Q q^{1/4} (e^{\pi i v} - e^{-\pi i v}) \prod_{\nu=1}^{\infty} (1 - q^{2\nu} e^{2\pi i v})(1 - q^{2\nu} e^{-2\pi i v}). \quad (4)$$

If in these formulae one puts $v = 0$, in the last after differentiating once, then

$$\begin{aligned}
 \vartheta_{00} &= Q \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1})^2, \\
 \vartheta_{01} &= Q \prod_{\nu=1}^{\infty} (1 - q^{2\nu-1})^2, \\
 \vartheta_{10} &= 2Qq^{1/4} \prod_{\nu=1}^{\infty} (1 + q^{2\nu})^2, \\
 \vartheta'_{11} &= 2\pi Qq^{1/4} \prod_{\nu=1}^{\infty} (1 - q^{2\nu})^2.
 \end{aligned} \tag{5}$$

By this, using § 23, (5),

$$\vartheta'_{11} = \pi \vartheta_{00} \vartheta_{01} \vartheta_{10},$$

the factor Q can be determined. First one obtains

$$1 = Q^2 \prod_{\nu=1}^{\infty} \frac{(1 - q^{2\nu-1})^2 (1 + q^{2\nu-1})^2 (1 + q^{2\nu})^2}{(1 - q^{2\nu})^2},$$

or, disposing of the still undetermined sign of Q , and thereby of the signs of the ϑ -functions,

$$Q = \prod_{\nu=1}^{\infty} \frac{1 - q^{2\nu}}{(1 + q^{2\nu})(1 + q^{2\nu-1})(1 - q^{2\nu-1})}.$$

In the denominator one can put, in place of $\prod(1 + q^{2\nu})(1 + q^{2\nu-1})$, the product $\prod(1 + q^\nu)$, and the numerator $\prod(1 - q^{2\nu})$ can be decomposed as

$$\prod(1 - q^\nu)(1 + q^\nu) = \prod(1 - q^{2\nu})(1 - q^{2\nu-1})(1 + q^\nu).$$

Thus finally

$$Q = \prod_{\nu=1}^{\infty} (1 - q^{2\nu}). \tag{6}$$

The expressions (1), (2), (3), (4) can be represented in real form when the multiplication of the conjugate imaginary factors is carried out. One obtains

$$\begin{aligned}
 \vartheta_{00}(v) &= \prod_{\nu=1}^{\infty} (1 - q^{2\nu})(1 + 2q^{2\nu-1} \cos 2\pi v + q^{4\nu-2}), \\
 \vartheta_{01}(v) &= \prod_{\nu=1}^{\infty} (1 - q^{2\nu})(1 - 2q^{2\nu-1} \cos 2\pi v + q^{4\nu-2}), \\
 \vartheta_{10}(v) &= 2q^{1/4} \cos \pi v \prod_{\nu=1}^{\infty} (1 - q^{2\nu})(1 + 2q^{2\nu} \cos 2\pi v + q^{4\nu}), \\
 \vartheta_{11}(v) &= 2q^{1/4} \sin \pi v \prod_{\nu=1}^{\infty} (1 - q^{2\nu})(1 - 2q^{2\nu} \cos 2\pi v + q^{4\nu}).
 \end{aligned} \tag{7}$$

If one introduces the expression (6) into the last equation (5), then

$$\vartheta'_{11} = 2\pi q^{1/4} \prod_{\nu=1}^{\infty} (1 - q^{2\nu})^3,$$

and if therefore

$$\eta(\omega) = q^{1/12} \prod_{\nu=1}^{\infty} (1 - q^{2\nu}), \tag{8}$$

then

$$\vartheta'_{11} = 2\pi\eta(\omega)^3. \quad (9)$$

Eliminating Q from (5) with the aid of the relation

$$Q = q^{-1/12}\eta(\omega),$$

we put

$$\vartheta_{00} = \eta(\omega)f(\omega)^2, \quad \vartheta_{01} = \eta(\omega)f_1(\omega)^2, \quad \vartheta_{10} = \eta(\omega)f_2(\omega)^2, \quad (10)$$

where the functions $f(\omega), f_1(\omega), f_2(\omega)$ are defined as follows:

$$\begin{aligned} f(\omega) &= q^{-1/24} \prod_{\nu=1}^{\infty} (1 + q^{2\nu-1}), \\ f_1(\omega) &= q^{-1/24} \prod_{\nu=1}^{\infty} (1 - q^{2\nu-1}), \\ f_2(\omega) &= \sqrt{2} q^{1/12} \prod_{\nu=1}^{\infty} (1 + q^{2\nu}). \end{aligned} \quad (11)$$

The functions $\eta(\omega), f(\omega), f_1(\omega), f_2(\omega)$ will play an important role in our later considerations. From § 21, (14), by (10), one gets the relation

$$f(\omega)^8 = f_1(\omega)^8 + f_2(\omega)^8, \quad (12)$$

and from § 23, (5), by (9),

$$f(\omega)f_1(\omega)f_2(\omega) = \sqrt{2}. \quad (13)$$

The last formula also follows from (11) by use of the identical relation

$$\prod (1 + q^\nu)(1 - q^{2\nu-1}) = \prod \frac{(1 + q^\nu)(1 - q^\nu)}{1 - q^{2\nu}} = 1.$$

§ 25. Representation of the ϑ -functions by infinite series.

The second way of arriving at the representation of the ϑ -functions consists in seeking to form a convergent infinite series satisfying the fundamental equations.

First observe that, because of the condition

$$\vartheta_{00}(v+1) = \vartheta_{00}(v),$$

the function $\vartheta_{00}(v)$ can be regarded as a single-valued function of $e^{2\pi iv}$, and put accordingly

$$\vartheta_{00}(v) = \sum_{\nu=-\infty}^{\infty} A_\nu e^{2\pi i \nu v}.$$

Then the differential equation § 20, (4) gives for A_ν the condition

$$\frac{dA_\nu}{d\omega} = \pi i \nu^2 A_\nu, \quad A_\nu = c_\nu e^{\pi i \omega \nu^2} = c_\nu q^{\nu^2},$$

where c_ν is constant with respect to ω . Thus

$$\vartheta_{00}(v) = \sum c_\nu q^{\nu^2} e^{2\pi i \nu v},$$

and hence

$$\vartheta_{00}(v + \omega) = \sum c_\nu q^{\nu^2 + 2\nu} e^{2\pi i \nu v} = q^{-1} e^{-2\pi i v} \sum c_\nu q^{(\nu+1)^2} e^{2\pi i (\nu+1)v}.$$

Since ν runs from $-\infty$ to $+\infty$, it is permissible in this formula to put $\nu - 1$ in place of ν ; doing this gives

$$\vartheta_{00}(v + \omega) = q^{-1} e^{-2\pi i v} \sum c_{\nu-1} q^{\nu^2} e^{2\pi i \nu v}.$$

According to the second of the fundamental equations § 20, (1), the two series

$$\sum c_{\nu} q^{\nu^2} e^{2\pi i \nu v} \quad \text{and} \quad \sum c_{\nu-1} q^{\nu^2} e^{2\pi i \nu v}$$

must therefore agree; that is,

$$c_{\nu-1} = c_{\nu}.$$

All the c_{ν} are consequently equal to one another; that they have the value 1 follows by comparing the two developments

$$\sum_{\nu=-\infty}^{\infty} q^{\nu^2} e^{2\pi i \nu v}, \quad \prod_{\nu=1}^{\infty} (1 - q^{2\nu})(1 + q^{2\nu-1} e^{2\pi i v})(1 + q^{2\nu-1} e^{-2\pi i v})$$

for the value $q = 0$.

The development thus found for $\vartheta_{00}(v)$ can also be written in both forms:

$$\begin{aligned} \vartheta_{00}(v) &= \sum_{\nu=-\infty}^{\infty} q^{\nu^2} e^{2\pi i \nu v} \\ &= 1 + 2q \cos 2\pi v + 2q^4 \cos 4\pi v + 2q^9 \cos 6\pi v + \cdots, \end{aligned} \quad (1)$$

and from it, by § 21, (8), through increasing v by

$$\frac{1}{2}, \quad \frac{\omega}{2}, \quad \frac{1+\omega}{2},$$

one obtains the developments for the three remaining ϑ -functions:

$$\begin{aligned} \vartheta_{01}(v) &= \sum_{\nu=-\infty}^{\infty} (-1)^{\nu} q^{\nu^2} e^{2\pi i \nu v} \\ &= 1 - 2q \cos 2\pi v + 2q^4 \cos 4\pi v - 2q^9 \cos 6\pi v + \cdots, \end{aligned} \quad (2)$$

$$\begin{aligned} \vartheta_{10}(v) &= \sum_{\nu=-\infty}^{\infty} q^{(2\nu+1)^2/4} e^{(2\nu+1)\pi i v} \\ &= 2q^{1/4} \cos \pi v + 2q^{9/4} \cos 3\pi v + 2q^{25/4} \cos 5\pi v + \cdots, \end{aligned} \quad (3)$$

$$\begin{aligned} \vartheta_{11}(v) &= -i \sum_{\nu=-\infty}^{\infty} (-1)^{\nu} q^{(2\nu+1)^2/4} e^{(2\nu+1)\pi i v} \\ &= 2q^{1/4} \sin \pi v - 2q^{9/4} \sin 3\pi v + 2q^{25/4} \sin 5\pi v + \cdots. \end{aligned} \quad (4)$$

For later applications we draw the following conclusions from these developments:

If the imaginary part of ω grows without bound, so that the absolute value of q vanishes, then

$$\begin{aligned} \vartheta_{00}(v) &= 1, & \vartheta_{01}(v) &= 1, & q^{-1/4} \vartheta_{10}(v) &= 2 \cos \pi v, \\ q^{-1/4} \vartheta_{11}(v) &= 2 \sin \pi v. \end{aligned}$$

Assume ω purely imaginary, hence q real, positive, and a proper fraction. Then for real v :

1. $\vartheta_{00}(v), \vartheta_{01}(v), \vartheta_{10}(v), \vartheta_{11}(v)$ are real, and if v lies between 0 and $1/2$, positive by § 24, (7); consequently also $\vartheta_{00}, \vartheta_{01}, \vartheta_{10}, \vartheta'_{11}$ are positive.

2. $\vartheta_{00}(iv), \vartheta_{01}(iv), \vartheta_{10}(iv), -i\vartheta_{11}(iv)$ are real, and so long as v lies between 0 and $-i\omega/2$, positive.

3. $\vartheta_{00}(\frac{1}{2} + iv)$, $\vartheta_{01}(\frac{1}{2} + iv)$, $i\vartheta_{10}(\frac{1}{2} + iv)$, $\vartheta_{11}(\frac{1}{2} + iv)$ are real, and so long as v lies between 0 and $-i\omega/2$, positive.

4. The functions

$$q^{1/4}e^{\pi iv}\vartheta_{00}\left(\frac{\omega}{2} + v\right), \quad -iq^{1/4}e^{\pi iv}\vartheta_{01}\left(\frac{\omega}{2} + v\right),$$

$$q^{1/4}e^{\pi iv}\vartheta_{10}\left(\frac{\omega}{2} + v\right), \quad -iq^{1/4}e^{\pi iv}\vartheta_{11}\left(\frac{\omega}{2} + v\right)$$

are real, and so long as v lies between 0 and $1/2$, positive.

§ 26. Development of ϑ -quotients.

If v denotes a variable and a an arbitrary constant, then the quotients

$$\frac{\vartheta_{g_1, g_2}(v + a)}{\vartheta_{g'_1, g'_2}(v)}$$

are doubly periodic functions of the second kind and first order with periodicity factors $(-1)^{g_1+g'_1}$, $(-1)^{g_2+g'_2}e^{-2\pi ia}$. If one appends an exponential factor $e^{\lambda v}$, then λ and a can be so determined that the periodicity factors become arbitrary prescribed quantities. Taking the two main characteristics (g_1, g_2) , (g'_1, g'_2) in all possible ways, one obtains sixteen such functions. We start from one among them, for which, appending a constant factor, we choose

$$F = \frac{\vartheta'_{11}\vartheta_{11}(v + a)}{2\pi i \vartheta_{11}(v)\vartheta_{11}(a)}. \quad (1)$$

This is a single-valued function $F(z)$ of the variable

$$z = e^{2\pi iv},$$

and, if as before $q = e^{\pi i\omega}$, it has the periodicity property

$$F(q^2 z) = e^{-2\pi ia} F(z), \quad (2)$$

from which, for every integral ν ,

$$F(q^{2\nu} z) = e^{-2\pi i\nu a} F(z).$$

The function $F(z)$ becomes infinite for a finite z only when one of the factors $q^{2\nu} z - 1$ vanishes, and in particular

$$(z - 1)F(z) = 1 \quad (z = 1). \quad (3)$$

But by (2)

$$(q^{2\nu} z - 1)F(q^{2\nu} z) = e^{-2\pi i\nu a} F(z)(q^{2\nu} z - 1),$$

and consequently, by (3),

$$F(z)(q^{2\nu} z - 1) = e^{2\pi i\nu a} \quad (z = q^{-2\nu}). \quad (4)$$

Put, under the supposition of convergence still to be examined more closely,

$$S(z) = \sum_{\nu=-\infty}^{\infty} \frac{e^{2\pi i\nu a}}{q^{2\nu} z - 1}. \quad (5)$$

Then, replacing z by $q^2 z$ and ν by $\nu - 1$, one obtains

$$S(q^2 z) = e^{-2\pi ia} S(z), \quad (6)$$

and the difference $F(z) - S(z)$, considered as a function of v , would therefore by (3) and (4) be an entire doubly periodic function of the second kind. Such a function must, however, by § 16, 6, be an exponential function; hence

$$F(z) = S(z) + Ce^{-2\pi imv},$$

where m is an integer and C a constant. From (2) and (6), however, if $C \neq 0$, it follows that

$$e^{-2\pi im\omega} = e^{-2\pi ia};$$

thus $a = m\omega + n$, where m, n are integers. But if a is a period, $F(z)$ reduces to a constant or an exponential function. Otherwise $C = 0$, and the development follows:

$$\frac{\vartheta'_{11}\vartheta_{11}(v+a)}{2\pi i \vartheta_{11}(v)\vartheta_{11}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{e^{2\pi i\nu a}}{q^{2\nu}e^{2\pi i\nu} - 1}. \quad (7)$$

We have presupposed the convergence of this series for all values of v . To judge it, observe that the general term of this series becomes, for $\nu = +\infty$, equal to $e^{2\pi i\nu a}$, and for $\nu = -\infty$, equal to $e^{-2\pi i\nu a}e^{-2\pi i\nu(\omega-a)}$. Thus convergence will take place when the imaginary parts of a and of $\omega - a$ are positive. Hence, if

$$\omega = \omega' + i\omega'', \quad a = a' + ia'',$$

the condition of convergence is

$$0 < a'' < \omega''. \quad (8)$$

If in (7) one replaces a by $a + \frac{1}{2}$, then one obtains the development for a second of the sixteen functions:

$$\frac{\vartheta'_{11}\vartheta_{10}(v+a)}{2\pi i \vartheta_{11}(v)\vartheta_{10}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{(-1)^\nu e^{2\pi i\nu a}}{q^{2\nu}e^{2\pi i\nu} - 1}. \quad (9)$$

If in (7) and (9) one changes a into $a + \frac{1}{2}\omega$, then

$$\frac{\vartheta'_{11}\vartheta_{01}(v+a)}{2\pi i \vartheta_{11}(v)\vartheta_{01}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{e^{2\pi i\nu a} q^\nu e^{\pi i\nu}}{q^{2\nu}e^{2\pi i\nu} - 1}, \quad (10)$$

$$\frac{\vartheta'_{11}\vartheta_{00}(v+a)}{2\pi i \vartheta_{11}(v)\vartheta_{00}(a)} = \sum_{\nu=-\infty}^{\infty} \frac{(-1)^\nu e^{2\pi i\nu a} q^\nu e^{\pi i\nu}}{q^{2\nu}e^{2\pi i\nu} - 1}, \quad (11)$$

where it is to be noted, however, that in the last two formulae the condition of convergence is changed, namely to

$$-\frac{\omega''}{2} < a'' < \frac{\omega''}{2}.$$

From the four formulae the remaining twelve result if v is replaced by $v + \frac{1}{2}$, $v + \frac{1}{2}\omega$, $v + \frac{1}{2}(1 + \omega)$; the regions of convergence are not further changed. In this way the formulae of Table I at the end of the volume are derived.

The series thus obtained do not all converge for real values of a ; for example, the series (7), in which the part running toward the side of positive ν ,

$$\sum_{\nu=1}^{\infty} \frac{e^{2\pi i\nu a}}{q^{2\nu}e^{2\pi i\nu} - 1}, \quad (12)$$

ceases to converge when the imaginary part of a becomes equal to zero, while the other part remains convergent even then. But one obtains from (12) an expression which still converges for a real a if one adds the sum

$$\sum_{\nu=1}^{\infty} e^{-2\pi i\nu a} = -\frac{e^{-2\pi ia}}{e^{-2\pi ia} - 1}. \quad (13)$$

It thereby goes over into

$$\sum_{\nu=1}^{\infty} \frac{e^{2\pi i \nu a} q^{2\nu} e^{2\pi i \nu}}{q^{2\nu} e^{2\pi i \nu} - 1},$$

and this series remains convergent so long as the imaginary part a'' of a lies between $-\omega''$ and $+\omega''$.

Accordingly, from (7), if we separate off the term corresponding to the value $\nu = 0$ and replace the negative ν by $-\nu$, one obtains

$$\frac{\vartheta'_{11}(v+a)}{\pi \vartheta_{11}(v) \vartheta_{11}(a)} = \frac{2i}{e^{2\pi i v} - 1} + \frac{2ie^{2\pi i a}}{e^{2\pi i a} - 1} + 2i \sum_{\nu=1}^{\infty} \frac{e^{-2\pi i \nu a}}{q^{-2\nu} e^{2\pi i \nu} - 1} + 2i \sum_{\nu=1}^{\infty} \frac{q^{2\nu} e^{2\pi i \nu a} e^{2\pi i \nu}}{q^{2\nu} e^{2\pi i \nu} - 1}.$$

But

$$\frac{2i}{e^{2\pi i v} - 1} = \frac{e^{-\pi i v}}{\sin \pi v} = \cotg \pi v - i, \quad \frac{2ie^{2\pi i a}}{e^{2\pi i a} - 1} = \frac{e^{\pi i a}}{\sin \pi a} = \cotg \pi a + i,$$

and consequently this development can also be written

$$\frac{\vartheta'_{11}(v+a)}{\pi \vartheta_{11}(v) \vartheta_{11}(a)} = \cotg \pi v + \cotg \pi a - 2i \sum_{m=2,4,6,\dots} \left(\frac{q^m e^{m\pi i a}}{e^{-2\pi i v} - q^m} - \frac{q^m e^{-m\pi i a}}{e^{2\pi i v} - q^m} \right), \quad (14)$$

where m runs through the series of even numbers

$$m = 2, 4, 6, 8, \dots$$

The domain of validity of this development is

$$-\omega'' < a'' < \omega''.$$

In Table II these sixteen developments are assembled.

From these formulae three others are derivable by increasing v , or a , or both by $1/2$. But three others, which correspond to increasing v and a by $\omega/2$, must be derived directly. They show real form when q, a, v are real.

A third kind of development, in which the symmetry of the functions with respect to the two variables v, a comes to expression, is obtained by expanding, according to powers of q , the fractions that occurred in the preceding developments. Thus one obtains

$$\frac{q^m}{e^{-2\pi i v} - q^m} = \sum_{\nu=1}^{\infty} q^{m\nu} e^{2\pi i \nu v}, \quad \frac{q^m}{e^{2\pi i v} - q^m} = \sum_{\nu=1}^{\infty} q^{m\nu} e^{-2\pi i \nu v},$$

and these developments are valid as long as the absolute value of $qe^{2\pi i v}$ is a proper fraction, that is, if $v = v' + iv''$, so long as

$$-\omega'' < v'' < \omega''.$$

Thus from (14) there results

$$\frac{\vartheta'_{11}(v+a)}{\pi \vartheta_{11}(v) \vartheta_{11}(a)} = \cotg \pi v + \cotg \pi a + 4 \sum q^{mm'/2} \sin(ma + m'v)\pi,$$

where m, m' independently run through the even numbers

$$2, 4, 6, 8, \dots$$

This formula is valid for real a and v and, beyond that, as long as the imaginary part both of v and of a is in absolute value smaller than ω'' .

In Table III the sixteen formulae of this kind are assembled.²

²Such developments were considered by Jacobi, "sur la rotation d'un corps" (collected works, vol. II), then by Hermite, Annales de l'école normale, 1885, and by Kronecker, Berlin Academy, 1885. Compare also the Strasbourg dissertation of L. Vockerodt, 1905.